

Thm (Global Continuity):

let (X, d_X) , (Y, d_Y) be metric spaces, $f \in Y^X$ (i.e. $f: X \rightarrow Y$)

Then TFAE:

- (i) f is ctc on X
- (ii) If $V \subseteq Y$ is open, then $f^{-1}(V)$ is open
- (iii). If $G \subseteq Y$ is closed, then $f^{-1}(G)$ is closed.
- (iv). Whenever $\lim_{n \rightarrow \infty} x_n = x$ in X , then $\lim_{n \rightarrow \infty} f(x_n) = f(x)$ in Y .

proof: (i) $\Rightarrow$ (ii). If $V \subseteq Y$ is open, then V is nbh of all points in V . By previous thm, $f^{-1}(V)$ is nbh of any points in it. Hence $f^{-1}(V)$ is open.

(ii) $\Rightarrow$ (i). If $\forall V \subseteq Y$, we have $f^{-1}(V)$ is open, for x in X , and let N be a nbh of $f(x)$ in Y . Then $x \in f^{-1}(N)$ is a nbh of x . By the previous thm, done.

(ii) $\Leftrightarrow$ (iii). If $A \subseteq Y$, then $X \setminus f^{-1}(A) = f^{-1}(Y \setminus A)$.

If $A=U$ is open, then $Y \setminus A$ is closed.

If $A=G$ is closed, then $Y \setminus G$ is open.

Mix these, we are done.

(i) $\Leftrightarrow$ (iv) is just (i) $\Leftrightarrow$ (ii) applied to arbitrary x in X .
 $\square$.

Relativization: let (X, d_X) be a metric space.

$A \in \mathcal{P}(X) \setminus \{\emptyset\}$. Define the relativized metric

$$d_A: A \times A \rightarrow [0, \infty), \text{ by } d_A = d_X|_{A \times A}$$

We call (A, d_A) the relativized metric space (relative to (X, d_X)).

Prop'n: For (X, d_X) , A as above, we have for $V \subseteq A$ that

V is open in $(A, d_A) \Leftrightarrow$ there is open U in (X, d_X) s.t. $V = A \cap U$.

Proof: let $x \in A$, $\varepsilon > 0$. Write $B_X(x, \varepsilon)$ - usual open ball in (X, d_X)

$$B_A(x, \varepsilon) = \{y \in A : d_A(x, y) = d_X(x, y) < \varepsilon\} - \text{open ball in } (A, d_A)$$

Note that $B_A(x, \varepsilon) = B_X(x, \varepsilon) \cap A$.

($\Rightarrow$): If $x \in V$, there is $\varepsilon_x > 0$ s.t. $B_A(x, \varepsilon_x) \subseteq V$

$$\text{Thus, } V = \bigcup_{x \in V} \{x\} \subseteq \bigcup_{x \in V} B_A(x, \varepsilon_x) \subseteq V$$

$$\text{i.e. } V = \bigcup_{x \in V} B_A(x, \varepsilon_x). \text{ Let } U = \bigcup_{x \in V} B_X(x, \varepsilon_x)$$

$$\text{Then } U \cap A = \bigcup_{x \in V} B_X(x, \varepsilon_x) \cap A = \bigcup_{x \in V} B_A(x, \varepsilon_x) = V.$$

($\Leftarrow$): If $x \in A \cap U$, there is $\varepsilon > 0$ s.t. $B_X(x, \varepsilon) \subseteq U$.

$$\text{Hence } B_A(x, \varepsilon) = \underbrace{B_X(x, \varepsilon) \cap A}_{\subseteq U} \subseteq U \cap A = V$$

i.e. V is open in (A, d_A) .

□.

Notation: We call $V \subseteq A$, where $A \subseteq X$, (X, d_X) metric space, and A has relative V , and satisfies the above proposition, relatively open.

A relatively closed set in A is analogue to above.

ex.: (i) $n \in \mathbb{N}$, $p \in [1, \infty]$, $e_i = (1, 0, \dots, 0)$

In $(\mathbb{R}^n, \|\cdot\|_\infty)$, consider $A = B(0, 1) \cap B(e_i, 1]$.

Then A is relatively open in $B[0,1]$

A is relatively closed in $B(0,1)$

(ii). Consider $\mathbb{R}$, usual metric, $d(x,y) = |x-y|$.

Let $G = \mathbb{Q} \cap [-\sqrt{2}, \sqrt{2}]$ — relatively closed

$= \mathbb{Q} \cap (-\sqrt{2}, \sqrt{2})$ — relatively open.

Thm: Let (X, d_X) , (Y, d_Y) be metric spaces. $D \in \mathcal{P}(X) \setminus \{\emptyset\}$, $f \in Y^D$

Then TFAE:

(i). f is continuous on D (relative metric from (X, d_X))

(ii). for any open $V \in Y$ $f^{-1}(V)$ is (relatively) open in D .

(iii). for any open $G \in Y$ $f^{-1}(G)$ is (relatively) closed in D .

(iv). If $\lim_{n \rightarrow \infty} x_n = x$ in D [$x_n, x \in D$], then $\lim_{n \rightarrow \infty} f(x_n) = f(x)$ in Y .

Completeness :

Def'n : Let (X, d) be a metric space. A sequence $(x_n)_{n=1}^{\infty} \in X^{\omega}$ is Cauchy if $\forall \varepsilon > 0, \exists n_{\varepsilon} \in \mathbb{N}$ s.t. $m, n \geq n_{\varepsilon} \Rightarrow d(x_m, x_n) < \varepsilon$.

Fact : (Easy Exercise)

If $(x_n)_{n=1}^{\infty} \in X^{\omega}$ converges (to x), then $(x_n)_{n=1}^{\infty}$ is Cauchy.

Def'n : We say that metric space (X, d) is complete if every Cauchy sequence admits a limit.

If $(V, \|\cdot\|)$ is a normed space, then it is complete if it is complete w.r.t. $d(x, y) = \|x - y\|$.

Moral : $(\mathbb{R}, |\cdot|)$, i.e., $d_{\mathbb{R}}(x, y) = |x - y|$, is complete.

We always borrow completeness of $\mathbb{R}$ to define/prove

completeness of (X, d)

$\Rightarrow$ "Real Analysis"

Ex: (i) $(\mathbb{R}^n, \|\cdot\|_p)$, $p \in [1, \infty)$, is complete.

proof: First, if $x \in \mathbb{R}^n$, $j=1, \dots, n$

$$j^{\text{th}} \text{-coordinate} \quad |x_j| = (|x_j|^p)^{\frac{1}{p}} \leq \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}$$

Hence if $(x^{(m)})_{m=1}^{\infty} \in (\mathbb{R}^n)^{\mathbb{N}}$ $[x^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)})]$

is a Cauchy sequence, then for $j=1, \dots, n$

for $l, m \in \mathbb{N}$, we have

$$|x_j^{(l)} - x_j^{(m)}| \leq \|x^{(l)} - x^{(m)}\|_p$$

Since $(x^{(m)})_{m=1}^{\infty}$ is Cauchy in $(\mathbb{R}, |\cdot|)$ (v)

Since $(\mathbb{R}, |\cdot|)$ is complete, $x_j^{(u)} = \lim_{m \rightarrow \infty} x_j^{(m)}$ exists $\forall m$

Note that $\lim_{m \rightarrow \infty} \|x^{(m)} - x^{(u)}\|_p = \lim_{m \rightarrow \infty} \left(\sum_{j=1}^n |x_j^{(m)} - x_j^{(u)}|^p \right)^{\frac{1}{p}}$

$$= \left(\sum_{j=1}^n \underbrace{\left(\lim_{m \rightarrow \infty} |x_j^{(m)} - x_j^{(u)}| \right)^p}_{=0} \right)^{\frac{1}{p}}$$

$$= 0$$

iii). let $X \neq \emptyset$. $\tau: X \times X \rightarrow \{0,1\} \subseteq [0,+\infty)$

$$\tau(x,y) = \begin{cases} 1, & x \neq y \\ 0, & x = y \end{cases} \quad (\text{trivial metric})$$

let $(x_n)_{n=1}^{\infty} \in X^{\mathbb{N}}$ be Cauchy in (X, τ)

If $\varepsilon \in (0,1]$, then $\exists n_{\varepsilon} \in \mathbb{N}$ s.t. $m, n \geq n_{\varepsilon} \Rightarrow \tau(x_n, x_m) < \varepsilon \Rightarrow$

$$\Rightarrow \tau(x_n, x_m) = 0$$

For $n \geq n_{\varepsilon}$, we have $\tau(x_n, x_{n_{\varepsilon}}) = 0$. So $x_n = x_{n_{\varepsilon}}$, $\forall n \geq n_{\varepsilon}$

$$\Rightarrow \lim_{n \rightarrow \infty} x_n = x_{n_{\varepsilon}}, \text{ i.e., ultimately constant.}$$

Convergence of function:

let $(X, d_X), (Y, d_Y)$ be metric spaces.

let $(f_n)_{n=1}^{\infty} \in (Y^X)^{\mathbb{N}}$

If $f \in Y^X$, we say that

- f_n converge to f pointwise if $\forall x \in X$,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \text{ in } (Y, d_Y)$$

- f_n converge to f uniformly if

$$\lim_{n \rightarrow \infty} \sup_{x \in X} d_Y(f_n(x), f(x)) = 0$$

Equivalently, given $\varepsilon > 0$, $\exists n_\varepsilon \in \mathbb{N}$ st. $n \geq n_\varepsilon \Rightarrow d_Y(f_n(x), f(x)) < \varepsilon$.

Notation: For (X, d_X) , (Y, d_Y) metric spaces.

$$\text{let } C(X, Y) = \{ f \in Y^X : f \text{ is cts} \}$$

[WARNING: Really we should write $C((X, d_X), (Y, d_Y))$]

 tell us what "cts" means.

Theorem: let (X, d_X) , (Y, d_Y) be metric spaces.

(i) If $(f_n)_{n=1}^\infty \in C(X, Y)^\mathbb{N}$ with f_n converging uniformly to f in Y^X ,
then $f \in C(X, Y)$.

(ii). let $(f_n)_{n=1}^\infty \in (Y^X)^\mathbb{N}$ be uniformly Cauchy

- (Y, d_Y) is complete.

Then there exists $f \in Y^X$ for which f_n converges to f uniformly.

(iii). If $(f_n)_{n=1}^\infty \in C(X, Y)^N$ is uniformly Cauchy, and (Y, d_Y) is complete.

then there exists $f \in C(X, Y)$ so f_n converges to f uniformly.

Proof: (iii) is immediate from (i) and (ii).

(i). We fix x in X , and verify cty (continuity) of f at x_0 .

Since x_0 was chosen arbitrarily, this implies that f is cts on X .

First, given $\varepsilon > 0$, let N in $\mathbb{N}$ be so $n \geq N \Rightarrow d_Y(f_n(x), f(x)) < \frac{\varepsilon}{3}$

(Unif. conv.), $\forall x \in X$.

Second, focus on x_0 , let $\delta > 0$ be so $x \in B_X(x_0, \delta) \Rightarrow f_N(x) \in B_Y(f_N(x_0), \frac{\varepsilon}{3})$

where N is as above. We borrow f_N 's cty.

Now if $x \in B_X(x_0, \delta)$,

$$\begin{aligned} d_Y(f(x), f(x_0)) &\leq d_Y(f(x), f_N(x)) + d_Y(f_N(x), f_N(x_0)) + d_Y(f_N(x_0), f(x_0)) \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \end{aligned}$$

Def'n: $(f_n)_{n=1}^{\infty}$ is uniformly Cauchy if given $\varepsilon > 0$,

$$\exists n_\varepsilon \in \mathbb{N} \text{ s.t. } m, n \geq n_\varepsilon \Rightarrow d_Y(f_m(x), f_n(x)) < \varepsilon, \forall x \in X.$$

(ii): It's immediate that if we fix $x \in X$.

then $(f_n(x))_{n=1}^{\infty} \in Y^{\mathbb{N}}$ is Cauchy in (Y, d_Y) .

By assumption of completeness, $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ exists.

Thus, we have defined $f \in Y^X$

[Now: f_n converges to f pointwise]

We must show that f_n converges to f uniformly.

To this end, let now for $\varepsilon > 0$, $n_\varepsilon \in \mathbb{N}$ be so $m, n \geq n_\varepsilon$

$$\Rightarrow d_Y(f_m(x), f_n(x)) < \frac{\varepsilon}{2}, \forall x \in X.$$

let's fix for the moment, $x \in X$.

Since $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ (pointwise conv) in (Y, d_Y) ,

there is $n_x \in \mathbb{N}$ s.t. $n \geq n_x \Rightarrow d_Y(f_n(x), f(x)) < \frac{\varepsilon}{2}$.

Hence if $n \geq n_\varepsilon$, we can simply let $m \geq n_x, n_\varepsilon$

$$\begin{aligned} \text{and } d_Y(f_n(x), f(x)) &\leq d_Y(f_n(x), f_m(x)) + d_Y(f_m(x), f(x)) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Note that the choice of n only depends on ε , not on x .

and hence is uniform over all x in X .

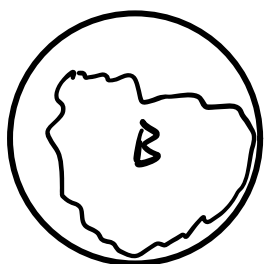
i.e. $f_n \rightarrow f$ uniformly.

□.

Def'n: (Bounded Set)

Let (X, d) be a metric space. A set $\emptyset \neq B \subseteq X$ is called bounded if $\text{diam}(B) = \sup_{x, y \in B} d(x, y) < \infty$

Remark: (i) If $x_0 \in B$ (above), then $B \subseteq B[x_0, \text{diam}(B)]$



[show that B is bounded $\Leftrightarrow$ for some $R > 0$,
 $B \subseteq B[x_0, R], x_0 \in B$]

(ii). if $x \in X$, then $B \subseteq B[X, \text{diam}(B) + d(x_0, x_1)]$, $x_1 \in B$

(iii). if B_1, B_2 are bounded, then so is $B_1 \cup B_2$

iv). WARNING: Boundedness depends on choices of metrics.

e.g. $\mathbb{R}$, $d(x, y) = |x - y|$, $\rho(x, y) = \frac{|x - y|}{1 + |x - y|}$

$\mathbb{R}$, itself, is bounded on $(\mathbb{R}, \rho)$!

Def'n: let $(X, d_X), (Y, d_Y)$ be metric spaces.

define $C_b(X, Y) = \{f \in Y^X : f \text{ is cts, fix } x \text{ bounded}\}$

WARNING: we should really write $C_b(X, d_X, Y, d_Y)$,
but we usually don't.]

Prop'n: let $(X, d_X), (Y, d_Y)$ be metric spaces.

If $f, g \in C_b(X, Y)$. let $d_c(f, g) = \sup_{x \in X} d_Y(f(x), g(x))$

Then d_c defines an metric on $C_b(X, Y)$

Proof: d_c takes $\mathbb{R}$ -values: $d_c(f, g) \leq \text{diam}(f(x) \cup g(x)) < \infty$

Non-negativity is obvious.

Non-degeneracy: $f \neq g \Rightarrow \exists x_0 \in X$ s.t. $f(x_0) \neq g(x_0)$.

Then $d_c(f, g) \geq d_Y(f(x_0), g(x_0)) > 0$

Subadditivity: let also, $h \in C_b(X, Y)$. Then

$$d_c(f, h) = \sup_{x \in X} d_Y(f(x), h(x))$$

$$\leq \sup_{x \in X} [d_Y(f(x), g(x)) + d_Y(g(x), h(x))]$$

$$\leq \sup_{x \in X} [d_Y(f(x), g(x)) + d_Y(g(x), h(x))]$$

$$= d_c(f, g) + d_c(g, h)$$

□.

Corollary: let (X, d_X) , (Y, d_Y) be metric spaces, with (Y, d_Y) complete.

Then $(C_b(X, Y), d_c)$ is complete.

proof: Notice that $\lim_{n \rightarrow \infty} d_c(f_n, f) = 0$ holds

$\Leftrightarrow f_n$ converges uniformly to f .

Use last Thm (ii)

□.

Notation: (X, d) metric space. let $C_b(X) = C_b(X, \mathbb{R})$,

$\mathbb{R}$ has usual metric $d(x, y) = |x - y|$

— vector space (pointwise, like $C[a, b]$)

— with norm $\|f\|_\infty = \sup_{x \in X} |f(x)|$

$(C_b(X), \|\cdot\|_\infty)$ is a complete normed space.

X — non-empty set.

$B_\infty(X) = \{f \in \mathbb{R}^X : f(x) \text{ is bounded}\}$

$\|f\|_\infty = \sup_{x \in X} |f(x)|$

Let complete normed space $(\ell_\infty(X), \|\cdot\|_\infty)$

$$\ell_\infty(U) = \ell_\infty$$

Lemma: let (X, d) be a metric space, and $(x_n)_{n=1}^\infty \in X^\mathbb{N}$

s.t. $\bullet (x_n)_{n=1}^\infty$ is Cauchy, and

$\bullet$ there are $n_1 < n_2 < n_3 < \dots$

s.t. $\lim_{k \rightarrow \infty} x_{n_k}$ exists in X .

Then $\lim_{n \rightarrow \infty} x_n = x$ exists in X .

proof: let $\varepsilon > 0$, $x = \lim_{k \rightarrow \infty} x_{n_k}$. Then we can find

$\bullet n_\varepsilon \in \mathbb{N}$ s.t. $m, n \geq n_\varepsilon \Rightarrow d(x_m, x_n) < \frac{\varepsilon}{2}$

$\bullet k_\varepsilon \in \mathbb{N}$ s.t. $k \geq k_\varepsilon \Rightarrow d(x_{n_k}, x) < \frac{\varepsilon}{2}$

If $n \geq n_\varepsilon$, then we can choose $k \geq k_\varepsilon$ s.t. $n_k \geq n$

(use that $n_1 < n_2 < \dots$), then

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

□.

Weierstrauss's M-Test : Applied to $C[a, b]$

Abstract M-Test : let $(V, \|\cdot\|)$ be a normed space. Then

$$(V, \|\cdot\|) \text{ is complete} \Leftrightarrow \forall (x_k)_{k=1}^{\infty} \in V^{\omega}, \\ \sum_{k=1}^{\infty} \|x_k\| < \infty \quad (\text{i.e. } \cup \mathcal{P})$$

$$\text{then } \sum_{k=1}^{\infty} x_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n x_k \text{ in } (V, \|\cdot\|)$$

Proof: ($\Rightarrow$) If $m < n$ in ω , then

$$\left\| \sum_{k=1}^m x_k - \sum_{k=1}^n x_k \right\| = \left\| \sum_{k=m+1}^n x_k \right\| \leq \sum_{k=m+1}^n \|x_k\|$$

and since $(\sum_{k=1}^n \|x_k\|)_{n=1}^{\infty} \cup \mathcal{P}$, the latter sequence in $\mathbb{R}$

is Cauchy, and $(\sum_{k=1}^n x_k)_{n=1}^{\infty}$ is Cauchy.

By the assumption of completeness,

$$\sum_{k=1}^{\infty} x_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n x_k, \text{ exists in } (V, \|\cdot\|)$$

($\Leftarrow$): let $n_1 \in \mathbb{N}$ st. $m, n > n_1 \Rightarrow \|x_m - x_n\| < \frac{1}{2}$

let $n_2 \in \mathbb{N}$ st. $n_2 > n_1$, and $m, n > n_2 \Rightarrow \|x_m - x_n\| < \frac{1}{2^2}$

$\vdots$

let $n_k \in \mathbb{N}$ st. $n_{k+1} > n_k$, and $m, n > n_{k+1} \Rightarrow \|x_m - x_n\| < \frac{1}{2^{k+1}}$

let $y_k = x_{n_{k+1}} - x_{n_k}$, so by design, $\|y_k\| < \frac{1}{2^k}$

Hence if $m < n$ in $\mathbb{N}$,

$$\sum_{k=m}^n \|y_k\| < \sum_{k=1}^m \frac{1}{2^k} = \frac{1}{2^m} \underbrace{\left(\frac{1}{2} + \frac{1}{2^2} + \dots \right)}_{< 1} < \frac{1}{2^{m-1}}$$

$\Rightarrow \sum_{k=1}^{\infty} \|y_k\|$ cjs in $\mathbb{R}$ as $\left(\sum_{k=1}^n \|y_k\| \right)$ is Cauchy.

$$\xrightarrow{\text{by assumption}} \sum_{k=1}^{\infty} y_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n y_k \text{ cjs in } (V, \|\cdot\|)$$

Now we have for $m \in \mathbb{N}$,

$$x_{n_m} = x_{n_1} + \sum_{k=1}^m (x_{n_{k+1}} - x_{n_k}) = x_{n_1} + \sum_{k=1}^m y_k$$

so $\lim_{m \rightarrow \infty} x_{n_m} = \lim_{m \rightarrow \infty} \left(x_{n_1} + \sum_{k=1}^m x_{n_k} \right)$ exists.

By last lemma, $(x_n)_{n=1}^{\infty}$ cjs in $(V, \|\cdot\|)$ $\square$.

Theorem: If $p \in [1, \infty)$, then $(\ell_p, \|\cdot\|_p)$ is complete.

proof: We use Abstract M-Test. Consider $(x^{(k)})_{k=1}^{\infty} \in \ell_p^{\mathbb{N}}$

$$\text{so } \sum_{k=1}^{\infty} \|x^{(k)}\|_p < \infty. \quad [\text{Each } x^{(k)} = (x_j^{(k)})_{j=1}^{\infty} \in \ell_p]$$

$$\text{Fix } j \in \mathbb{N}, \text{ so } |x_j^{(k)}| \leq \left(\sum_{i=1}^{\infty} |x_i^{(k)}|^p \right)^{\frac{1}{p}} = \|x^{(k)}\|_p$$

$$\text{and hence } \sum_{k=1}^n |x_j^{(k)}| \leq \sum_{k=1}^n \|x^{(k)}\|_p \leq \sum_{k=1}^{\infty} \|x^{(k)}\|_p < \infty.$$

$$\text{so } \sum_{k=1}^{\infty} |x_j^{(k)}| \text{ cvg in } (\mathbb{R}, |\cdot|)$$

Applying Abstract M-Test to $(\mathbb{R}, |\cdot|)$, we use

$$\text{that } x_j = \sum_{k=1}^{\infty} x_j^{(k)} \text{ cvg in } (\mathbb{R}, |\cdot|).$$

$$\text{let } x = (x_j)_{j=1}^{\infty} = \left(\sum_{k=1}^{\infty} x_j^{(k)} \right)_{j=1}^{\infty} \in \ell_p^{\mathbb{N}}$$

let's see that $x \in \ell_p$. Complete.

$$\|x\|_p = \left(\sum_{j=1}^{\infty} |x_j|^p \right)^{\frac{1}{p}} = \left(\sum_{j=1}^{\infty} \left| \sum_{k=1}^{\infty} x_j^{(k)} \right|^p \right)^{\frac{1}{p}}$$

$$= \left(\lim_{m \rightarrow \infty} \sum_{j=1}^m \left| \lim_{n \rightarrow \infty} \sum_{k=1}^n x_j^{(k)} \right|^p \right)^{\frac{1}{p}}$$

$$= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \left(\sum_{j=1}^m \left| \sum_{k=1}^n x_j^{(k)} \right|^p \right)^{\frac{1}{p}} \text{ (ctty of power fn)}$$

$$= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\sum_{j=1}^m |x_j^{(k)}|^p \right)^{\frac{1}{p}} \text{ (Minkowski)}$$

$$\leq \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \underbrace{\sum_{k=1}^n \|x^{(k)}\|_p}_{\text{indep of } m}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \|x^{(k)}\|_p = \sum_{k=1}^{\infty} \|x^{(k)}\|_p < \infty, \text{ by assumption.}$$

Now, we wish to see that $x = \lim_{n \rightarrow \infty} \sum_{k=1}^n x^{(k)}$

Complete $\|x - \sum_{k=1}^n x^{(k)}\|_p = \left\| \left(\sum_{k=n+1}^{\infty} x_j^{(k)} \right)_{j=1}^m \right\|_p$

$$\leq \sum_{k=n+1}^{\infty} \|x^{(k)}\|_p \xrightarrow{n \rightarrow \infty} 0, \text{ by assumption on } \sum x^{(k)}.$$

Conclusion: we found $x = \sum_{k=1}^{\infty} x^{(k)}$. By Abstract 1st-Test,

$(\ell_p, \|\cdot\|_p)$ is complete.

□.

Will Normed vector space always be complete?

No!

Example: (i), $V = \text{span} \{p_n : n \in \mathbb{N}\} \subset \mathcal{P}(\mathbb{R})$, $p_n \in \mathbb{R}^{[0,1]}$, $p_n(t) = t^n$

Fact: $(V, \|\cdot\|_p)$ is NOT complete.

(ii). let $1 < p < \infty$. let $(U, \|\cdot\|) = (\ell_1, \|\cdot\|_p)$, not complete. ($A \subset \mathbb{Q}$)

Ex: Consider $(C[0,2], \|\cdot\|_p)$, $p \in [1, +\infty)$

$$\text{let } n \in \mathbb{N}. \quad f_n(t) = \begin{cases} 0, & t \in [0, 1] \\ n(t-1), & t \in (1, 1+\frac{1}{n}) \\ 1, & t \in [1+\frac{1}{n}, 2] \end{cases}$$

$$\Rightarrow f_n \in C[0,2]$$

$$\begin{aligned} \text{If } m < n \text{ in } \mathbb{N}, \|f_m - f_n\| &= \left(\int_0^2 |f_m(t) - f_n(t)|^p \right)^{\frac{1}{p}} \\ &\leq \left(\int_1^{1+\frac{1}{n}} 1^p \right)^{\frac{1}{p}} = \frac{1}{n^{1/p}} \end{aligned}$$

Hence it follows that $(f_n)_{n=1}^{\infty}$ is Cauchy in $(C[0,2], \|\cdot\|_p)$

$$\text{let } h(t) = \begin{cases} 0, & t \in [0,1] \\ 1, & t \in (1,2] \end{cases} \Rightarrow \text{not cts.}$$

Easy to see $\lim_{n \rightarrow \infty} \|f_n - h\|_p = 0$.

let us suppose, for sake of contradiction,

that there were f in $C[0,2]$ s.t.

$$\lim_{n \rightarrow \infty} \|f_n - f\|_p = 0, \text{ i.e., } f = \lim_{n \rightarrow \infty} f_n \text{ in } (C[0,2], \|\cdot\|_p)$$

First, notice that

$$0 \leq \|f - h\|_p \leq \|f - f_n\|_p + \|f_n - h\|_p \rightarrow 0$$

$$\Rightarrow 0 = \|f - h\|_p$$

let $\delta \in (0,1)$ be so that $t \in (1-\delta, 1+\delta)$,

we have $|f(t) - f(1)| < \frac{1}{2}$

Then for $t \in (1-\delta, 1+\delta)$

$$|h(t) - f(t)| \geq |h(t) - f(1)| - \underbrace{|f(1) - f(t)|}_{> \frac{1}{2}}$$

$$\Rightarrow \|f - h\|_p^p = \int_0^2 |f - h|^p \geq \int_{1-\delta}^{1+\delta} |f - h|^p$$

$$\geq \delta \cdot \left(\inf_{t \in (1-\delta, 1+\delta)} |h(t) - f(1)| - \frac{1}{2} \right)^p > 0$$

~~*~~.

Fact. $(C[a,b], \|\cdot\|_p)$ NOT complete if $p \in [1, \infty)$

Lipschitz Function: let $(X, d_X), (Y, d_Y)$ be metric spaces.

A function $f: X \rightarrow Y$ is called Lipschitz if $\exists L > 0$ s.t.

$$d_Y(f(x), f(y)) \leq L \cdot d_X(x, y), \quad \forall x, y \in X.$$

$$\Rightarrow L \geq \frac{d_Y(f(x), f(y))}{d_X(x, y)}, \quad \forall x \neq y \in X.$$

Note that $\text{Lipschitz} \Rightarrow \underset{\text{cts}}{\text{uniformly}} \Rightarrow \text{cts}.$
 $\nLeftarrow \qquad \nLeftarrow$

Ex: (i) let $f: [0,1] \rightarrow \mathbb{R}$, $f(t) = \sqrt{t}$.

Fact: f is uniformly cts on $[0,1]$

($[0,1]$ is compact $\Rightarrow$ unif. cts. — visit this soon)

However, consider $s_n = \frac{1}{n+1}$, $t_n = \frac{1}{n}$ in $[0,1]$

then $\frac{\sqrt{t_n} - \sqrt{s_n}}{t_n - s_n} \xrightarrow{n \rightarrow \infty} \infty$, $\Rightarrow$ Not Lipschitz.

(ii). $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(t) = \cos(\pi t^2)$; so f is cts and bounded.

i.e. $f \in C_b(\mathbb{R})$. let $t_n = \sqrt{n}$, $s_n = \sqrt{\frac{n+1}{2}}$

so $|t_n - s_n| \xrightarrow{n \rightarrow \infty} 0$

but $|f(t_n) - f(s_n)| = 1 \Rightarrow$ not unif. cts.

Distance functions: let (X, d) be a metric space and $x_0 \in X$.

Define $f: X \rightarrow \mathbb{R}$, $f(x) = d(x, x_0)$.

Notice that for x, y in X , $d(x, x_0) \leq d(x, y) + d(y, x_0)$

$$\Rightarrow f(x) - f(y) = d(x, x_0) - d(y, x_0) \leq d(x, y)$$

Interchanging roles of x, y , we also see that

$$f(y) - f(x) \leq d(x, y)$$

$$\text{Hence } |f(x) - f(y)| \leq d(x, y)$$

$\Rightarrow$ It's Lipschitz with $L=1$.

Prop'n: If $x = \lim_{n \rightarrow \infty} x_n$, $y = \lim_{n \rightarrow \infty} y_n$ in (X, d)

$$\text{then } d(x, y) = \lim_{n \rightarrow \infty} d(x_n, y_n)$$

$$\begin{aligned} \text{proof: } 0 &\leq |d(x, y) - d(x_n, y_n)| = |d(x, y) - d(x, y_n) + d(x, y_n) - d(x_n, y_n)| \\ &\leq |d(x, y) - d(x, y_n)| + |d(x, y_n) - d(x_n, y_n)| \end{aligned}$$

$$\leq d(y, y_n) + d(x, x_n) \xrightarrow{n \rightarrow \infty} 0$$

• Example: (i) (Concrete)

$$\mathcal{L} = \left\{ x = (x_n)_{n=1}^{\infty} \in \mathbb{R}^{\mathbb{N}} : \text{for some } n_x \text{ in } \mathbb{N}, x_n = 0 \forall n > n_x \right\}$$

"finitely supported" sequences.

If $p \in [1, \infty)$, define for x in $\mathcal{L}$:

$$\|x\|_p = \left(\sum_{n=1}^{n_x} |x_n|^p \right)^{\frac{1}{p}}$$

Fact: $(\mathcal{L}, \|\cdot\|_p)$ NOT complete ($\mathbb{A} \geq \mathbb{Q}$)

• $\mathcal{L} \subseteq \ell_p$ and is dense in ℓ_p

and, by Thm, $(\ell_p, \|\cdot\|_p)$ is complete.

Also, $\mathcal{L} \hookrightarrow \ell_p$ is an isometry
inclusion

$(\ell_p, \|\cdot\|_p)$
is the
completion of
 $(\mathcal{L}, \|\cdot\|_p)$

(ii). (Abstract)

$a < b$ in $\mathbb{R}$, $(C[a, b], \|\cdot\|_p)$, $p \in [1, \infty)$

FACT: (A_3, Q_3) : completion exists:

$$(\overline{C}^p[a,b], \|\cdot\|_p)$$

Recall: If (X, d) is a metric space, $\emptyset \neq A \subseteq X$, then

$$\text{diam}(A) = \sup_{x,y \in A} d(x,y) \quad (\text{might be } \infty)$$

we are typically interested in finite diameter.

Remark : For A as above, $\text{diam}(\bar{A}) = \text{diam}(A)$
 $\quad \quad \quad \nwarrow$ closure.

First, $A \subseteq \bar{A} \Rightarrow \text{diam}(A) \leq \text{diam}(\bar{A})$

To see reverse inequality, let $x, y \in \bar{A}$, $\varepsilon > 0$.

By ball-meet characterization of $\bar{A}$, we can find

$$x' \in B(x, \frac{\varepsilon}{2}) \cap A, y' \in B(y, \frac{\varepsilon}{2}) \cap A. \text{ Then}$$

$$\begin{aligned}
d(x, y) &\leq d(x, x') + d(x', y') + d(y', y) \\
&< \frac{\varepsilon}{2} + \text{diam}(A) + \frac{\varepsilon}{2} \\
&= \text{diam}(A) + \varepsilon
\end{aligned}$$

(if $\text{diam}(A) < \infty$). Since $\varepsilon > 0$ is arbitrary, we see that

$$\text{diam}(\bar{A}) = \sup_{x, y \in \bar{A}} d(x, y) \leq \text{diam}(A)$$

Cantor's Nested Sequence Theorem

Let (X, d) be a metric space.

Then

(X, d) is complete $\Leftrightarrow$

for any sequence of sets

- $X \supseteq F_1 \supseteq F_2 \supseteq \dots$
- with each F_n closed, $F_n \neq \emptyset$
- $\lim_{n \rightarrow \infty} \text{diam}(F_n) = 0$

We have that $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$

proof: $(\Rightarrow)$: We let $F_n \subseteq X$, $\forall n \in \mathbb{N}$ satisfies the three conditions on the right.

let for each n in $\mathbb{N}$, $x_n \in F_n$

Then if $m < n$ in $\mathbb{N}$,

$$\bullet \quad x_m \in F_m, x_n \in F_n \subseteq F_m \Rightarrow x_m, x_n \in F_m$$

$$\bullet \quad d(x_m, x_n) \leq \text{diam}(F_m)$$

Thus $(x_n)_{n=1}^{\infty}$ is Cauchy. By assumption of completion,

$$x = \lim_{n \rightarrow \infty} x_n \text{ exists in } X.$$

We then note, for each m in $\mathbb{N}$, that

$$x \in F_m \text{ (as } \{x_n\}_{n=m}^{\infty} \subseteq F_m), \text{ and}$$

we use sequence limit characterization of closed set.

$$\text{Then we select } x \in \bigcap_{m=1}^{\infty} F_m, \text{ i.e. } \bigcap_{m=1}^{\infty} F_m \neq \emptyset.$$

($\Leftarrow$): let $(x_k)_{k=1}^{\infty}$ be a Cauchy sequence on X .

$$\text{Then let } F_n = \overline{\{x_n\}_{n=1}^{\infty}} \text{ (closure on } X)$$

Notice that $\cdot F_n$ is closed and non-empty

• if $m \leq n$, then $\{x_k\}_{k=m}^{\infty} \supseteq \{x_k\}_{k=n}^{\infty}$

So $F_m = \overline{\{x_k\}_{k=m}^{\infty}} \supseteq \overline{\{x_k\}_{k=n}^{\infty}} = F_n$

- $\text{diam}(I_n) = \text{diam}(\{X_k\}_{k=n}^{\infty})$ (by Remark)

$$\xrightarrow{n \rightarrow \infty} 0 \quad (\text{Cauchy})$$

Thus, by assumption $\bigcap_{n \geq 1} F_n \neq \emptyset$.

If $x \in \bigcap_{n=1}^{\infty} F_n$, then $x \in F_n$ for each n , so

$$d(x, x_n) \leq \text{diam}(F_n) \xrightarrow{n \rightarrow \infty} 0$$

D.

Example: ($\text{diam}(T_n) \xrightarrow{n \rightarrow \infty} 0$ is important)

In I_1 , let $e_n = (0, \dots, 0, 1, 0, \dots)$
 $\quad\quad\quad \uparrow$
 $\quad\quad\quad n^{\text{th}} \text{ position}.$

let for each $n \in \mathbb{N}$, $\bar{f}_n = \{e_k\}_{k=n}^{\infty}$

- Notice:
- each F_n is closed (check that $A \setminus F_n$ is open)
 - $\text{diam}(F_n) = 2$
 - $F_1 \supseteq F_2 \supseteq \dots$

But $\bigcap_{n=1}^{\infty} F_n = \emptyset$

// MIDTERM
CUTOFF .

Compactness:

Def'n: Let (X, d) be a metric space, $K \subseteq X$.

We say that K is compact if

- for every open cover $(\{U_i\}_{i \in I})$, each U_i open,
 $K \subseteq \bigcup_{i \in I} U_i$ for K .
- there exists a finite subcover

$$(\{U_{i_1}, \dots, U_{i_n}\} \text{ s.t. } K \subseteq \bigcup_{k=1}^n U_{i_k})$$

If $K = X$, itself is compact, we call (X, d)

a compact metric space

Relativization: If $K \subseteq X$, as above,

K is compact in $(X, d) \Leftrightarrow (K, d_K)$ (relativized metric)
is a compact metric space

Indeed, $\{U_i\}_{i \in I}$ open cover
of K in $(X, d) \Leftrightarrow \{K \cap U_i\}_{i \in I}$
is an open cover
of K in (K, d_K)

Theorem: (Finite Intersection Property)

Let (X, d) be a metric space. Then

(X, d) is a compact
metric space $\Leftrightarrow$ for every family of closed set $\{F_i\}_{i \in I}$
s.t. $\forall \{F_{i_1}, \dots, F_{i_n}\}, \bigcap_{k=1}^n F_{i_k} \neq \emptyset$ (f.i.p.)
we have that $\bigcap_{i \in I} F_i \neq \emptyset$.

Proof: ($\Rightarrow$): (Contradiction) Given a family $\{\bar{F}_i\}_{i \in I}$ of closed sets

with f.i.p, but with $\bigcap_{i \in I} \bar{F}_i = \emptyset$. Let for each $i \in I$

$$U_i = X \setminus \bar{F}_i, \text{ so}$$

$$\bigcup_{i \in I} U_i = \bigcup_{i \in I} (X \setminus \bar{F}_i) = X \setminus \bigcap_{i \in I} \bar{F}_i = X \setminus \emptyset = X$$

i.e. $\{U_i\}_{i \in I}$ is an open cover, since (X, d) is compact.

$$\Rightarrow \exists \text{ some } \{U_{i_1}, \dots, U_{i_n}\} \text{ s.t. } X = \bigcup_{k=1}^n U_{i_k}$$

$$\text{but, then, as above, we see that } \bigcap_{i=1}^n \bar{F}_i = \emptyset \quad \#$$

($\Leftarrow$): Similar. Start with open cover $\{U_i\}_{i \in I}$.

Take $\bar{F}_i = X \setminus U_i$. And then yield a similar $\#$.

□.

Proposition: If (X, d) is a compact metric space.

then (X, d) is complete.

proof: (Cantor's Nested Set Theorem)

let $F_1 \supseteq F_2 \supseteq \dots \neq \emptyset$, each F_n closed

with $\lim_{n \rightarrow \infty} \text{diam}(F_n) = 0$.

If $n_1 < n_2 < \dots < n_m$, then $\bigcap_{k=1}^m F_{k} = F_{n_m} \neq \emptyset$.

So $\{F_n\}_{n=1}^{\infty}$ satisfies f.i.p. By assumption on compactness.

and FIP Theorem tells us that $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$.

By Cantor's Nested Set Theorem, (X, d) is complete. □.

Corollary: let (X, d) be a metric space, $K \subseteq X$.

If K is compact on (X, d) , then K is closed.

proof: If $K = \emptyset$, then it is trivial.

If $K \neq \emptyset$, then $x \in \bar{K}$ (closure), then

$x = \lim_{n \rightarrow \infty} x_n$ for some $(x_n)_{n=1}^{\infty} \in K^{\mathbb{N}}$ (seq. of char. of closure)

Then, $(x_n)_{n=1}^{\infty}$ is Cauchy in (X, d) , hence

Cauchy in (K, d_K) , and thus has a limit in K .

By uniqueness of limits in (X, d) , we see $x \in K$

i.e., $K = \overline{K}$

□.

Prop'n: If (X, d) is a compact metric space,

$C \subseteq X$ is closed, then C is compact.

proof: let $\{U_i\}_{i \in I}$ be an open cover of C .

Then $\{U_i\}_{i \in I} \cup \{X \setminus C\}$ is an open cover of X .

By assumption we get finite subcover $\{U_{i_1}, \dots, U_{i_n}, X \setminus C\}$

hence $\{U_{i_1}, \dots, U_{i_n}\}$ is a cover of C

□.

Remark: (i). Finite sets are always compact.

(ii). Finite unions of compact sets are compact.

(iii). Arbitrary intersections of compact sets are compact.

Thm. If $a < b$ in $\mathbb{R}$, then $[a, b]$ is compact in $(\mathbb{R}, |\cdot|)$

proof: let $\{U_i\}_{i \in I}$ be an open cover of $[a, b]$

let $S = \{x \in [a, b] : [a, x] \text{ admits a finite subcover of } \{U_i\}_{i \in I}\}$

Notice that $[a, a] = \{a\}$, then $a \in S$ (extract one interior of $\{U_i\}_{i \in I}$)

So $S \neq \emptyset$. Since $S \subseteq [a, b]$, S is bounded.

$s = \sup S$ exists.

Suppose, for $\#$, that $s < b$. Consider U_{i_0} s.t.

$s \in U_{i_0}$. There exists $\varepsilon > 0$ (say also $\varepsilon < b - s$).

so $(s - \varepsilon, s + \varepsilon) \subseteq U_{i_0}$

By def'n of $s = \sup S$, then $\exists x \in (s - \varepsilon, s] \cap S$

Then there exists $\{u_1, \dots, u_n\}$ s.t.

$$[a, x] \subseteq \bigcup_{k=1}^n u_k, \text{ Then } [a, s + \frac{\varepsilon}{2}] \subseteq \bigcup_{k=1}^n u_k$$

So $s + \frac{\varepsilon}{2} \in S$, $\nexists$ to $s = \sup S$.

□.

Ex: $e_n = \underbrace{\{0, \dots, 0, 1, 0, \dots\}}_n \in \mathcal{A}_1$

then $F_n = \{e_k\}_{k=n}^{\infty} \subset B[0, 1] \subset \mathcal{A}_1$

- closed
- bounded

$\hookrightarrow$ closed unit ball
centered at 0

$$F_1 \supseteq F_2 \supseteq \dots \Rightarrow \text{f.i.p.} \left. \vphantom{\begin{matrix} F_1 \\ F_2 \end{matrix}} \right\} \Rightarrow B[0, 1] \text{ NOT}$$

But $\bigcap_{n=1}^{\infty} F_n = \emptyset$ COMPLETE!

Def'n: let (X, d) be a metric space, $K \subseteq X$.

We say that K is totally bounded if

$$\cdot \forall \varepsilon > 0, \exists x_1, \dots, x_n \text{ in } K \text{ so } K \subseteq \bigcup_{k=1}^n B(x_k, \varepsilon)$$

Remark: If $\varepsilon > 0$, we may consider $\varepsilon' \in (0, \varepsilon)$, and the

definition of total boundedness provides $x_1, \dots, x_n$ in K .

$$\text{so } K \subseteq \bigcup_{k=1}^n B(x_k, \varepsilon') \subseteq \bigcup_{k=1}^n B(x_k, \varepsilon)$$

Thus, the total boundedness is equivalent to

being able to cover K with finitely many open balls
of arbitrary radius $\varepsilon > 0$.

FACT: totally bounded $\Rightarrow$ bounded
"finite diameter"

Ex: In ℓ_1 , $e_n = (0, \dots, 0, 1, 0, \dots)$. let $F = \{e_n\}_{n=1}^{\infty} \subset \ell_1$

Then $\text{diam}(F) = 2$, therefore F is bounded

However, $F \subseteq \bigcup_{n=1}^{\infty} B[\varepsilon_n, r]$, $\forall \varepsilon \in (0, 1)$

and no subcollection, in particular finite subcollection of $\{B[\varepsilon_n, r]\}_{n=1}^{\infty}$, covers F .

Theorem (Characterization of Compactness)

Let (X, d) be a metric space. Then TFAE:

- (i) (X, d) is compact
- (ii) (X, d) is complete and totally bounded.
- (iii) any sequence $(x_n)_{n=1}^{\infty} \in X^{\mathbb{N}}$ admits a subsequence which converges inside of X .

Proof: (i) $\Rightarrow$ (ii) We already saw that a compact metric space is complete (F.I.P. & Nested Set Thm)

If $\varepsilon > 0$, then $X = \bigcup_{i \in X} B(x, \varepsilon)$

and by compactness, there are $x_1, \dots, x_n$ s.t.

$$X = \bigcup_{k=1}^n B(x_k, \varepsilon) = \bigcup_{k=1}^n B[x_k, \varepsilon]$$

(ii) $\Rightarrow$ (iii): let $(x_n)_{n=1}^{\infty} \in X^{\mathbb{N}}$. By assumption of total boundedness, for each n in $\mathbb{N}$, there is a finite collection of closed balls

$B_{n1}, \dots, B_{nm_n}$ s.t. each ball has radius $\frac{1}{n}$, $X = \bigcup_{k=1}^{m_n} B_{nk}$.

Let $\cdot B_1$ be chosen from $B_{11}, \dots, B_{1m_1}$ so

$$|\{n \in \mathbb{N} : x_n \in B_1\}| = \aleph_0 \quad (\text{"Pigeonhole"})$$

$\cdot B_2$ be chosen from $B_{21}, \dots, B_{2m_2}$ so

$$|\{n \in \mathbb{N} : x_n \in B_1 \cap B_2\}| = \aleph_0$$

$\vdots$

$\cdot B_k$ be chosen from $B_{k1}, \dots, B_{km_k}$ so

$$|\{n \in \mathbb{N} : x_n \in \bigcap_{j=1}^k B_j\}| = \aleph_0$$

For each n in $\mathbb{N}$. let $F_n = \bigcap_{k=1}^n B_k$.

$\cdot$ each F_n is closed

$\cdot$ each $F_n \neq \emptyset$, $F_n \cap \{x_d\}_{d=1}^{\infty} \neq \emptyset$.

- $F_1 \supseteq F_2 \supseteq \dots$

- $\text{diam}(F_n) \leq \text{diam}(B_n) = \frac{2}{n} \xrightarrow{n \rightarrow \infty} 0$

So our assumption of completeness and Cantor's Nested Set Thm.

provides $x_0 \in \bigcap_{n=1}^{\infty} F_n$

Now we construct a subsequence $(x_{n_k})_{k=1}^{\infty}$, let

- $n_1 = \min \{n \in \mathbb{N} : x_n \in F_1\}$

- $n_2 = \min \{n \in \mathbb{N} : n > n_1, x_n \in F_2\}$

⋮

- $n_{k+1} = \min \{n \in \mathbb{N} : n > n_k, x_n \in F_{k+1}\}$

Then for k in $\mathbb{N}$, $x_0, x_{n_k} \in F_k$, so

$$d(x_0, x_{n_k}) \leq \text{diam}(F_k) \leq \frac{2}{k} \xrightarrow{k \rightarrow \infty} 0$$

So $x_0 = \lim_{k \rightarrow \infty} x_{n_k}$.

(iii) $\Rightarrow$ (i): (A) let's show that (iii) $\Rightarrow (X, d)$ is totally bounded.

Contrapositive proof: Assume (X, d) is not totally bounded.

Thus, there exists $\varepsilon > 0$ st. no finite collection of balls

$$B[x_1, \varepsilon], \dots, B[x_n, \varepsilon] \text{ has } \bigcup_{k=1}^n B[x_k, \varepsilon] \subsetneq X.$$

Then proceed as follows:

- pick $x_1 \in X$
- pick $x_2 \in X \setminus B[x_1, \varepsilon]$
- pick $x_3 \in X \setminus \bigcup_{k=1}^2 B[x_k, \varepsilon]$
- $\vdots$
- pick $x_n \in X \setminus \bigcup_{k=1}^{n-1} B[x_k, \varepsilon]$

So, we get a sequence $(x_n)_{n=1}^{\infty}$. Notice

if $m \neq n$, say $m < n$, then

$$x_n \in X \setminus B[x_m, \varepsilon], \text{ so } d(x_m, x_n) > \varepsilon.$$

Hence if we take any subsequence $(x_{n_k})_{k=1}^{\infty}$ of $(x_n)_{n=1}^{\infty}$

we would have, for $h \neq l$, that $d(x_{n_h}, x_{n_l}) > \varepsilon$

so $(x_{n_k})_{k=1}^{\infty}$ is not Cauchy, and hence not convergent.

(B), We show, under assumption of (iii), that a given open cover $\{U_i\}_{i \in I}$ of X , there exists $r > 0$ s.t. $\forall x \in X$, we have $B(x, r) \subseteq U_{i_x}$ for some $i_x \in I$.

Contradiction proof: We assume (iii), and also assume that $\forall r > 0$,

$$\exists x \in X \text{ s.t. } B(x, r) \not\subseteq U_i, \forall i \in I \quad (f)$$

Thus, by (f), $\forall n \in \mathbb{N}$, $\exists x_n \in X$ s.t. $B(x_n, \frac{1}{n}) \not\subseteq U_i, \forall i \in I$ (*)

Thus we get a choice of x_n for each n , i.e. $(x_n)_{n=1}^{\infty} \in X^{\mathbb{N}}$.

By assumption, $\exists$ a subsequence which converges.

i.e., $\lim_{k \rightarrow \infty} x_{n_k} = x_0$ in (X, d) . Then there is $i_0 \in I$

so $x_0 \in U_{i_0}$, and further there is $\varepsilon > 0$ s.t. $B(x_0, \varepsilon) \subseteq U_{i_0}$.

Given that $\varepsilon > 0$, let k_ε be so $k \geq k_\varepsilon \Rightarrow d(x_{n_k}, x_0) < \frac{\varepsilon}{2}$

Then, for $k \geq k_\varepsilon$ s.t. $\frac{1}{n_k} < \frac{\varepsilon}{2}$, we have for $x \in B(x_{n_k}, \frac{1}{n_k})$ that

$$d(x, x_0) \leq d(x, x_{n_k}) + d(x_{n_k}, x_0)$$

$$< \frac{1}{n_k} + \frac{\varepsilon}{2} < \varepsilon$$

But this contradicts to (*), and hence (f) cannot hold.

(C) We assume (iii). Then given an open cover $\{U_i\}_{i \in I}$ of X .

we let $r > 0$ be provided as in (B). Then (A) provided that (X, d) is totally bounded.

$$\text{so } X = \bigcup_{k=1}^n B(x_k, r), \quad x_1, \dots, x_n \in X.$$

But, by (B), each x_n admits $i_n \in I$.

so $B(x_n, r) \subseteq U_{i_n}$. Hence

$$X = \bigcup_{k=1}^n B(x_k, r) \subseteq \bigcup_{i=1}^n U_{i_n} \subseteq X$$

$$\text{so } \bigcup_{k=1}^n B(x_k, r) = \bigcup_{i=1}^n U_{i_n}, \text{ i.e., } \{U_{i_1}, \dots, U_{i_n}\} \text{ is finite subcover}$$

Hence (iii) $\Rightarrow$ (i)

□.

Remark: Consider $\mathbb{R}^n$. We saw (A2) all norms $\|\cdot\|_p, 1 \leq p \leq \infty$.

are equivalent. Again by A2, notions of open sets are

the same for all of these norms.

Hence, each norm produces same notion of compact sets.

Theorem (i) (BW Thm): If $R > 0$ in $\mathbb{R}$, then $[-R, R]^n$ is compact in $\mathbb{R}^n$.

(ii). (HB Thm): If $K \subseteq \mathbb{R}^n$, K compact $\Leftrightarrow K$ is closed & bounded.

proof: (i). Induction on n .

Base case: $n=1$; already saw that $[-R, R]$ is compact in $\mathbb{R}$ (Earlier, $[a, b] \subseteq \mathbb{R}$ is compact)

Induction step: Assume the result is true $\forall n \geq 1$.

we let $(x^{(m)})_{m=1}^{\infty} \subseteq [-R, R]^{n+1}$ be a sequence.

$$\text{In } [-R, R]^{n+1} = [-R, R]^n \times [-R, R]$$

$$\begin{aligned} \text{write each } x^{(m)} &= (x_1^{(m)}, \dots, x_n^{(m)}, x_{n+1}^{(m)}) \\ &= (y^{(m)}, t^{(m)}) \text{ , } y^{(m)} \in [-R, R]^n, \text{ } t^{(m)} \in \mathbb{R}. \end{aligned}$$

By the induction hypothesis on n , $(y^{(m)})_{m=1}^{\infty} \in [-R, R]^n$ admits

a convergent subsequence $(y^{(m_k)})_{k=1}^{\infty}$ with $y = \lim_{k \rightarrow \infty} y^{(m_k)}$

in $[-R, R]^n$. Then the sequence $(t^{(m)})_{m=1}^{\infty}$ in $[-R, R]$

admits a w.p. subseq $(t^{(m_k)})_{k=1}^{\infty}$ with $t = \lim_{k \rightarrow \infty} t^{(m_k)}$

in $[-R, R]^n$.

$$\text{Then } (y, t) = \lim_{l \rightarrow \infty} (y^{(m_{k_l})}, t^{(m_{k_l})})$$

$$= \lim_{l \rightarrow \infty} x^{(m_{k_l})}$$

i.e., $(x^{(m)})_{m=1}^{\infty}$ admits a w.p. subseq.

$\Rightarrow [-R, R]^{n+1}$ is compact.

(ii). ($\Rightarrow$): K compact $\Rightarrow$ complete $\Rightarrow$ closed
 $\Rightarrow$ totally bounded $\Rightarrow$ bounded.

($\Leftarrow$): K bounded $\Rightarrow \exists R > 0$ s.t. $K \subseteq B_{\infty}[0, R] = [-R, R]^n$
 $\|\cdot\|_{\infty}$ ball $\rightarrow$

we saw in (i) that $[0, 1]^n$ is compact.

so K is closed subset of a compact subset

$\Rightarrow K$ is compact.

Example: $K = \prod_{k=1}^{\infty} \{0, \frac{1}{2^k}\} = \{x = (x_k)_{k=1}^{\infty}, \text{ each } x_k \in \{0, \frac{1}{2^k}\}\} \subset \mathcal{L}_1$

Fact: K is closed in $\mathcal{L}_1(A_3 \mathcal{Q}_1, k' = k)$.

and since $(\mathcal{L}_1, \|\cdot\|_1)$ is complete, K is complete in $(\mathcal{L}_1, \|\cdot\|_1)$
(A3 Q2)

let's verify K is totally bounded, hence compact.

let $\varepsilon > 0$, find $n \in \mathbb{N}$ s.t. $\frac{1}{2^n} < \varepsilon$. For $b_1, \dots, b_n \in \{0, 1\}$,

$$\text{let } x_{b_1 \dots b_n} = \sum_{k=1}^n \frac{b_k}{2^k} e_k = \left(\frac{b_1}{2}, \frac{b_2}{2^2}, \dots, \frac{b_n}{2^n}, 0, 0, \dots \right)$$

so $\{x_{b_1 \dots b_n}\}_{(b_1 \dots b_n) \in \{0, 1\}^n}$ is a set of size 2^n .

if $x \in K$, then $x = (\frac{b_1}{2}, \dots, \frac{b_n}{2^n}, \dots)$, $b_k \in \{0, 1\}$

let $b_1, \dots, b_n$ be the first n entries of this rep'n.

$$\text{then } \|x - x_{b_1 \dots b_n}\|_1 = \|(0, \dots, 0, \frac{b_{n+1}}{2^{n+1}}, \dots)\|_1$$

$$= \sum_{k=n+1}^{\infty} \left| \frac{b_k}{2^k} \right|$$

$$= \sum_{k=n+1}^{\infty} \frac{b_k}{2^k}$$

$$\leq \sum_{k=n+1}^{\infty} \frac{1}{2^k} = \frac{1}{2^n}$$

$$\text{i.e., } B[x_{b_1 \dots b_n}, \frac{1}{2^n}] \subseteq B[x_{b_1 \dots b_n}, \varepsilon]$$

$$\Rightarrow K \subseteq \bigcup_{(b_1 \dots b_n) \in \{0,1\}^n} B[x_{b_1 \dots b_n}, \varepsilon]$$



finite union of closed set $\Rightarrow$ closed

$\Rightarrow K$ is bounded.

Theorem: let (X, d_X) be compact metric space, (Y, d_Y) any metric

$f: X \rightarrow Y$ be cts fun (i.e., $f \in C(X, Y)$)

Then $f(X) = \{f(x) : x \in X\}$ is compact in (Y, d_Y)

[Hence $f \in C_b(X, Y)$, i.e., $f(x)$ totally bdd $\Rightarrow$ bdd]

Proof: let $\{U_i\}_{i \in I}$ be an open cover of $f(X)$.

Then, for each i in I , $U_i = f^{-1}(U_i)$ is open in X .

(Since f is cts)

Then $\{U_i\}_{i \in I}$ is an open cover of X , hence admitting

finite subcovers $\{U_{i_1}, \dots, U_{i_n}\}$. Thus,

$\{U_{i_1}, \dots, U_{i_n}\}$ is a finite subcover of $f(X)$

□.

Corollary: (EVT) If (X, d) is a compact metric space.

$f: X \rightarrow \mathbb{R}$ is cts, then $\exists x_{\min}, x_{\max} \in X$ s.t.

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}), \quad \forall x \in X.$$

proof: $f(X)$ is compact (last thm), hence is closed & bounded (HB thm)

let $m = \inf f(X)$, $M = \sup f(X)$. For each $n \in \mathbb{N}$,

let $x_n, y_n \in X$ be so

$$f(x_n) \in [m, m + \frac{1}{n}), \quad f(y_n) \in (M - \frac{1}{n}, M]$$

By inf, sup $\Rightarrow x_{\min} = \lim_{n \rightarrow \infty} x_{n_k}$, $x_{\max} = \lim_{n \rightarrow \infty} y_{n_k}$

for suitable sub seqs. By cty, $f(x_{\min}) = \lim_{n \rightarrow \infty} f(x_{n_k}) = m$

similarly, $f(x_{\max}) = M$

□.

Prop'n: (Seq char. of unif. cty)

let $(X, d_X), (Y, d_Y)$ be metric spaces.

$f: X \rightarrow Y$. Then

f is unif. cts $\Leftrightarrow$ whenever $d_X(x_n, y_n) \xrightarrow{n \rightarrow \infty} 0$
we have $d_Y(f(x_n), f(y_n)) \xrightarrow{n \rightarrow \infty} 0$

($\Rightarrow$): Given $\varepsilon > 0$, $\delta > 0$ s.t. $d_X(x, y) < \delta$ in $X \Rightarrow d_Y(f(x), f(y)) < \varepsilon$ in Y .

If $d_X(x_n, y_n) \xrightarrow{n \rightarrow \infty} 0$ in X , let $n_f \in \mathbb{N}$ s.t. $n \geq n_f \Rightarrow d_X(x_n, y_n) < \delta$

Then if $n \geq n_f$, $d_Y(f(x_n), f(y_n)) < \varepsilon$, i.e. $d_Y(f(x_n), f(y_n)) \xrightarrow{n \rightarrow \infty} 0$

($\Leftarrow$): Contrapositive: Assume f is not unif. cts.

$\Rightarrow \exists \varepsilon_0 > 0$ s.t. $\forall \delta > 0$, $\exists x, y \in X$ s.t. $d_X(x, y) < \delta$

but $d_Y(f(x), f(y)) \geq \varepsilon_0$.

Now for $n \in \mathbb{N}$, let x_n, y_n be so

$d_X(x_n, y_n) < \frac{1}{n}$ but $d_Y(f(x_n), f(y_n)) \geq \varepsilon_0$

Then $\lim_{n \rightarrow \infty} d_X(x_n, y_n) = 0$ (Squeeze) but

$\lim_{n \rightarrow \infty} (f(x_n), f(y_n)) \neq 0$ (if exists) $\#$.

Thm: let (X, d_X) be compact metric space, (Y, d_Y) any metric space, and $f \in C(X, Y) = C_b(X, Y)$

Then f is unif. cts.

proof: Sp. $(x_n)_{n=1}^{\infty}, (y_n)_{n=1}^{\infty}$ in X^{∞} satisfy $\lim_{n \rightarrow \infty} d_Y(x_n, y_n) = 0$,

but $\exists \varepsilon_0 > 0$ s.t. $d_Y(f(x_n), f(y_n)) \geq \varepsilon_0$

we find • a cys subseq

$(x_k)_{k=1}^{\infty}$ w/ $x_0 = \lim_{k \rightarrow \infty} x_{n_k}$ in X .

• a cys subseq

$(y_{n_{k_l}})_{l=1}^{\infty}$ w/ $y_0 = \lim_{l \rightarrow \infty} y_{n_{k_l}}$

Notice that $d_X(x_0, y_0) = \lim_{l \rightarrow \infty} d_X(x_{n_{k_l}}, y_{n_{k_l}}) = 0$

$\Rightarrow x_0 = y_0$. Hence cty of f

$$f(x_0) = \lim_{k \rightarrow \infty} f(x_{n_k}) = f(y_0)$$

$\Rightarrow d_Y(f(x_0), f(y_0)) = 0$, But each $d_Y(f(x_{n_k}), f(y_{n_k})) \geq \varepsilon > 0$
 $\nexists$.

Theorem: Let V be a finite dimensional $\mathbb{R}$ -vector space.

Then any two norms on V are equivalent ($\approx$)

proof: Let $\{v_1, \dots, v_n\}$ be a basis for V . We define

a norm on V by

$$\|x\| = \sum_{j=1}^n |x_j| \quad \text{where } x = \sum_{j=1}^n x_j v_j, \text{ each } x_j \in \mathbb{R}.$$

[easy to check it's a norm]

Let $\|\cdot\|$ be any (other) norm on V . We'll show that

$$\|\cdot\| \approx \|\cdot\| \text{ (norm above), hence any third norm } \approx \|\cdot\|$$

Then any two norms are equivalent.

First, if $x \in V$, $x = \sum_{j=1}^n x_j u_j$, each $x_j \in \mathbb{R}$.

$$\|x\| \leq \sum_{j=1}^n |x_j| \cdot \|u_j\| \leq M \cdot \sum_{j=1}^n |x_j| = M \cdot \|x\| \quad (*)$$

where $M = \max_{j=1, \dots, n} \|u_j\|$. $\Rightarrow \|\cdot\|$ is Lipschitz $\Rightarrow$ is cts

Define $\iota: V \rightarrow \mathbb{R}^n$ by $\iota(x) = (x_1, \dots, x_n)$

where $x = \sum_{j=1}^n x_j u_j$. Then ι is an isometry between

$(V, \|\cdot\|)$ and $(\mathbb{R}^n, \|\cdot\|_1)$, so $\iota^{-1}: \mathbb{R}^n \rightarrow V$ is also

an isometry, hence cts, but

$$S_1 = \{y \in \mathbb{R}^n : \|y\|_1 = 1\} \quad \text{"sphere"}$$

which is closed (inverse image of $\{1\} \subseteq \mathbb{R}$ by cts fcn

$\|\cdot\|_1$ on $\mathbb{R}^n$), and is bounded ($S_1 \subseteq B_1[0,1] \subset \mathbb{R}^n$)

Hence, by Heine-Brown Thm, S_1 is compact

Thus, $\iota^{-1}(S_1) = K$, is compact in $(V, \|\cdot\|)$

Thus, by EVT, there is $x_{\min}$ in K_1 , so $\|x_{\min}\| \leq \|x\|, \forall x \in K_1$.

Since $0 \notin K_1$ (as $0 \notin S_1$, and ι^{-1} is injective & linear)

as $\|\cdot\|$ is a norm. Note that $K_1 = \{x \in V : \underbrace{\|x\|}_{\geq 1} \geq 1\}$ (ι isometry)

so if $x \in V \setminus \{0\}$, we see that $\frac{1}{\|x\|} x \in K_1$.

and hence $\|\frac{1}{\|x\|} x\| \geq \|x_{\min}\| = m$

$$\text{So } \|x\| \geq m \|x\| \quad (**)$$

Combining (*) and (**), we have

$$m \|x\| \leq \|x\| \leq M \cdot \|x\|, \quad \forall x \in V.$$

$$\text{i.e. } \|\cdot\| \approx \|\cdot\|$$

□.

Remark: A finite dimensional $\mathbb{R}$ -v.s. has only one equiv class

of norms. So def'n of open, closed etc. sets is

indep of choice of norm. (A₂)

Corollary: let $(V, \|\cdot\|)$ be a f. dim normed vector space.

(i). (HB) A subset $K \subseteq V$ is compact

$\Leftrightarrow K$ is closed & bounded.

(ii). $(V, \|\cdot\|)$ is complete.

proof: (i). we may assume $\|\cdot\|$ is as last proof.

$\Rightarrow \exists$ isometry $\iota: (V, \|\cdot\|) \rightarrow (\mathbb{R}^n, \|\cdot\|_1)$

K bounded $\Rightarrow \iota(K)$ bounded in $\mathbb{R}^n$. (easy)

K closed $\Rightarrow \iota(K)$ is closed

(let $(y^{(n)})_{n=1}^{\infty} \subset \iota(K)^{\text{int}}$ w/ $y^{(n)} \xrightarrow{n \rightarrow \infty} y$.

then $\iota^{-1}(y) = \lim_{n \rightarrow \infty} \iota^{-1}(y^{(n)}) \in K$ closed)

so $y = \iota(\iota^{-1}(y)) \in \iota(K)$. By HB on $\mathbb{R}^n$,

$\iota(K)$ is compact so $K = \iota^{-1}(\iota(K))$ is compact.

(ii) If $(x^{(n)})_{n=1}^{\infty}$ is Cauchy in $(V, \|\cdot\|)$, then

$\{x^{(n)}\}_{n=1}^{\infty}$ is bounded, so $K = \overline{\{x^{(n)}\}_{n=1}^{\infty}}$ is compact.

But K is complete. So $(x^{(n)})_{n=1}^{\infty}$ admits a limit in K . $\square$

Proposition: let $(V, \|\cdot\|_V)$, $(W, \|\cdot\|_W)$ be f. dim. v.s.

Then if $T: V \rightarrow W$ is lin, then T is cts.

proof: let $\{v_1, \dots, v_n\}$, $\{w_1, \dots, w_m\}$ be bases for V, W .

Then for each $i \in \{1, \dots, n\}$, we have

$$Tv_i = \sum_{j=1}^m t_{ij} w_j, \quad t_{ij} \in \mathbb{R}.$$

Thus if $x = \sum_{i=1}^n x_i v_i$, each $x_i \in \mathbb{R}$.

$$\text{we have } Tx = \sum_{i=1}^n x_i Tv_i = \sum_{i=1}^n \sum_{j=1}^m x_i t_{ij} w_j$$

let $\|\cdot\|$ on V be given by $\|x\| = \sum_{i=1}^n |x_i|$ (as before)

Notice that if $x^{(k)} \xrightarrow{k \rightarrow \infty} x$ in V , then $x_i^{(k)} \xrightarrow{k \rightarrow \infty} x_i$ for each $i=1, \dots, n$. Then

$$\begin{aligned} \|T x^{(k)} - T x\|_W &= \|T(x^{(k)} - x)\|_W = \left\| \sum_{i=1}^n \sum_{j=1}^m (x_i^{(k)} - x_i) t_{ij} w_j \right\|_W \\ &\leq \sum_{i=1}^n \sum_{j=1}^m |x_i^{(k)} - x_i| \cdot |t_{ij}| \cdot \|w_j\|_W \\ &\xrightarrow{k \rightarrow \infty} 0 \end{aligned}$$

$\Rightarrow T x^{(k)} \xrightarrow{k \rightarrow \infty} T x$, i.e., T is cts. $\square$.

Prop'n: let $(V, \|\cdot\|_V)$, $(W, \|\cdot\|_W)$ be normed spaces,

and let $T: V \rightarrow W$ is linear. Then TFAE.

(i). T is cts

(ii). $\|T\| = \sup \{ \|Tu\|_W : u \in \underbrace{B_V[0,1]}_{\text{closed unit radius ball around } 0 \text{ in } V} \} < \infty$

closed unit radius ball
around 0 in V .

$$(iii): \|Tv\|_W \leq \|T\| \cdot \|v\|_V, \quad \forall v \in V.$$

$$(iii') \quad T \text{ is Lipschitz. } \|Tv - Tv'\|_W \leq \|T\| \cdot \|v - v'\|_V$$

proof: $(i) \Rightarrow (i')$. T is cts, hence cts on 0 .

$$\Rightarrow \exists \delta > 0 \text{ s.t. } T(B_V(0, \delta)) \subseteq B_W(0, 1)$$

$$\text{If } v \in B_V(0, 1), \text{ then } \frac{\delta}{2} v \in B_V(0, \delta)$$

$$\text{so } \|Tv\|_W = \frac{2}{\delta} \|T(\frac{\delta}{2} v)\| \leq \frac{2}{\delta}$$

$$\text{Hence } \|T\| = \sup \{ \|Tv\|_W : v \in B_V(0, 1) \} < \infty$$

$$(i) \Rightarrow (ii): \text{ If } v \in V \setminus \{0\}, \quad \frac{1}{\|v\|_V} v \in B_V(0, 1) \text{ . so}$$

$$\|Tv\| = \|v\|_V \cdot \|T(\frac{1}{\|v\|_V} v)\|$$

$$\leq \|v\|_V \cdot \|T\|$$

$$(ii') \Rightarrow (ii): \text{ If } v, v' \in V, \text{ then } \|Tv - Tv'\|_W = \|T(v - v')\|_W \leq \|T\| \cdot \|v - v'\|_V$$

$$(ii') \Rightarrow (i): \text{ Trivial.}$$

Remark: (i) $B(V, W) = \{T \in W^V : T \text{ is linear and cts}\}$

is a $\mathbb{R}$ -linear space with pointwise operations

• then $\|\cdot\|$ is a norm on $B(V, W)$

(ii). If, additionally, $(W, \|\cdot\|_W)$ is complete, then $(B(V, W), \|\cdot\|)$

is a complete normed space

(Similar to $(C_b(X), \|\cdot\|_\infty)$ is complete)

Examples: Consider $C[0,1] = \{f \in \mathbb{R}^{[0,1]} : f \text{ cts}\}$

Note: Since $[0,1]$ is compact, each $f \in C[0,1]$ is bounded

i.e., $f([0,1]) \subset \mathbb{R}$ is compact.

Let $\varphi: C[0,1] \rightarrow \mathbb{R}$ by $\varphi(f) = f(\frac{1}{2})$ (evaluation at $\frac{1}{2}$)

Fact: φ is linear.

(i). Consider $(C[0,1], \|\cdot\|_\infty)$. If $f \in C[0,1]$, then

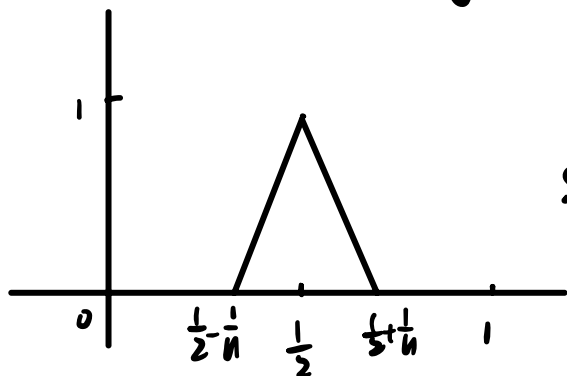
$$|\varphi(f)| = |f(\frac{1}{2})| \leq \max_{t \in [0,1]} |f(t)| = \|f\|_{\infty} \quad \text{by EVT, max instead of sup}$$

$$\Rightarrow \|\varphi\| \leq 1. \text{ In fact, } \|\varphi\| = 1$$

$$(ii) \text{ Now let } p \in [1, +\infty) \text{ so for } f \in C[0,1], \|f\|_p = (\int_0^1 |f|^p)^{\frac{1}{p}}$$

We will show that φ is not cts $(C[0,1], \|\cdot\|_p)$

$$\text{Let } f_n(t) = \begin{cases} n(t - \frac{1}{2} + \frac{1}{n}), & t \in [\frac{1}{2} - \frac{1}{n}, \frac{1}{2}] \\ n(\frac{1}{2} + \frac{1}{n} - t), & t \in [\frac{1}{2}, \frac{1}{2} + \frac{1}{n}] \\ 0, & \text{otherwise.} \end{cases}$$



so $\forall n \in \mathbb{N}$, $f_n(t)$ is cts, $f_n(\frac{1}{2}) = 1$

$$\text{Now } \|f_n\|_p = (\int_0^1 |f_n(t)|^p dt)^{\frac{1}{p}}$$

$$= (\int_{\frac{1}{2}-\frac{1}{n}}^{\frac{1}{2}+\frac{1}{n}} \underbrace{f_n(t)^p}_{\leq 1} dt)^{\frac{1}{p}}$$

$$\leq (\int_{\frac{1}{2}-\frac{1}{n}}^{\frac{1}{2}+\frac{1}{n}} 1^p dt)^{\frac{1}{p}} = (\frac{2}{n})^{\frac{1}{p}} = \frac{2^{\frac{1}{p}}}{n^{\frac{1}{p}}} \xrightarrow{n \rightarrow \infty} 0$$

Hence we cannot find $L > 0$ s.t.

$$l = |\varphi(f_n)| \leq L \cdot \|f_n\|_p \xrightarrow{n \rightarrow \infty} 0$$

i.e., φ is not Lipschitz $\Rightarrow \varphi$ not cts.

MORAL: Pointwise evaluation is not cts on $(C[0,1], \|\cdot\|_p)$
 $p \in [1, \infty)$

So we define $L_p[0,1]$ to be the completion, we cannot expect

to have a linear function $\varphi: L_p[0,1] \rightarrow \mathbb{R}$ with

$$\varphi(f) = f(\tfrac{1}{2}) \text{ for } f \in [0,1]$$

Contraction Mapping Theorem (CMT)

Let (X, d) be a complete metric space, and $T: X \rightarrow X$

be a strict contraction, i.e., $\exists c \in [0,1)$ s.t.

$$d(Tx, Ty) \leq c \cdot d(x, y) \text{ (Lipschitz with constraints } < 1)$$

Then (i) $\exists!$ $x_{f,x}$ in X s.t. $T(x_{f,x}) = x_{f,x}$, called a fixed point

(ii) $\forall x_0 \in X$, the sequence defined recursively by

$$x_n = T(x_{n-1}), \quad n \in \mathbb{N}$$

satisfies

$$d(x_n, x_{f,x}) \leq \frac{c^n}{1-c} d(T(x_0), x_0) \quad \text{constraint}$$

and it satisfies that $x_{f,x} = \lim_{n \rightarrow \infty} x_n$

proof: (ii). Let $x_0 \in X$, $(x_n)_{n=1}^{\infty}$ defined as above.

$$d(x_2, x_1) = d(T(x_1), T(x_0)) \leq c \cdot d(x_1, x_0) = c \cdot d(T(x_0), x_0)$$

Hence we want to see that

$$(*) \quad d(x_n, x_{n-1}) \leq c^{n-1} d(T(x_0), x_0)$$

The case $n=1$ is given. If $(*)$ is true for n ,

$$\begin{aligned} d(x_{n+1}, x_n) &= d(T(x_n), T(x_{n-1})) \\ &\leq c \cdot d(x_n, x_{n-1}) \\ &\leq c \cdot c^{n-1} \cdot d(T(x_0), x_0) = c^n \cdot d(T(x_0), x_0) \end{aligned}$$

So by induction, $(*)$ is true.

Now if $m < n$, then (*) tells us

$$\begin{aligned}
 d(x_n, x_m) &\leq \sum_{j=m}^{n-1} d(x_{j+1}, x_j) \quad (n-m \text{ times of } \Delta\text{-ineq}) \\
 &\leq \sum_{j=m}^{n-1} c^j d(Tx_0, x_0) \\
 &= c^m \cdot d(Tx_0, x_0) \cdot \sum_{j=0}^{n-m-1} c^j \\
 &\leq c^m \cdot d(Tx_0, x_0) \cdot \sum_{j=0}^{\infty} c^j = \frac{c^m}{1-c} d(Tx_0, x_0)
 \end{aligned}$$

and, as $c \xrightarrow{n \rightarrow \infty} 0$, we see that $(x_n)_{n=1}^{\infty}$ is Cauchy

in (X, d) . Then let $x_{f,x} = \lim_{n \rightarrow \infty} x_n$, which exists as

(X, d) is complete. Since T is Lipschitz, it's cts.

$$\text{So } x_{f,x} = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} T(x_n) = T(\lim_{n \rightarrow \infty} x_n) = T(x_{f,x})$$

$$\text{and further } d(x_{f,x}, x_m) = \lim_{n \rightarrow \infty} d(x_n, x_m)$$

$$\leq \lim_{n \rightarrow \infty} \frac{c^m}{1-c} d(Tx_0, x_0)$$

$$= \frac{c^m}{1-c} d(Tx_0, x_0)$$

Now we show uniqueness in (i).

If also $T(y) = y$, $y \in X$, then

$$d(x_{f,x}, y) = d(T(x_{f,x}), T(y)) \leq \underbrace{L}_{< 1} \cdot d(x_{f,x}, y) \rightarrow 0$$

Example: Assumption $d(T(x), T(y)) < d(x, y)$, $x, y \in X$

not sufficient to find fixed point.

(proof rely on geometric series)

Let $T: \underbrace{[1, +\infty)}_{\text{closed in } \mathbb{R}} \rightarrow \mathbb{R}$, $T(x) = x + \frac{1}{x}$

closed in $\mathbb{R}$.

so it's complete.

Sps we had a fixed point: $x_{f,x} \in [0, +\infty)$ s.t. $T(x_{f,x}) = x_{f,x}$

$$\text{Then } x_{f,x} = T(x_{f,x}) = x_{f,x} + \frac{1}{x_{f,x}} \Rightarrow 0 = \frac{1}{x_{f,x}} \quad \#$$